CHAPTER 6

Construction

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Figure 6.1: *Tehri dam in Uttarakhand*

According to the Ministry of Jal Shakti and the National Register of Large Dams (NRLD), India built over 6,138 dams between Independence and 2025. These dams are used for providing irrigation for agriculture, generating electricity, flood control and drinking water. One of the highest dams in India is the Tehri dam, with a height of 261 m built across the Bhagirathi river in Uttarakhand (Figure 6.1).

In this chapter, you will

Prepare a technical drawing of a structure you will construct	Prepare a bill of materials	Mark the drawing on the field
Prepare the foundation	Lay bricks to construct the selected structure	Complete the structure with curing, plastering, finishing and painting

6.1 Introduction

Construction simply means the process of building structures that people use in their daily lives. It enables us to live safely, travel and work comfortably, and protect ourselves from heat, rain and other weather conditions.

Since the beginning of human life, people have always needed a safe place to live. In early times, humans took shelter in caves. As they began to settle in one place and grow food, they started making simple homes using mud, grass, wood and stones. Slowly, as basic construction principles evolved, people learnt to use bricks, cement, iron and steel, which helped them build stronger and longer-lasting houses.

As villages turned into towns and towns into cities, the need for more houses and better facilities increased. Due to the lack of open space in cities, people started constructing multi-storey buildings. Along with houses, other structures, such as schools, hospitals, shops, factories, roads, bridges and railway stations, also became necessary. This is how the construction sector became an important part of everyday life, contributing to the nation's progress.

Today, construction uses modern tools, machines and materials, along with traditional skills. In this chapter, you will explore how structures are built through different materials and the steps involved in construction.

DID YOU KNOW?

Basic elements of building construction

Figure 6.2 shows the basic structural elements, that is, the different parts of any building.

Finishing works, such as plumbing, electrical wiring, plastering and painting are also carried out to complete the structure.

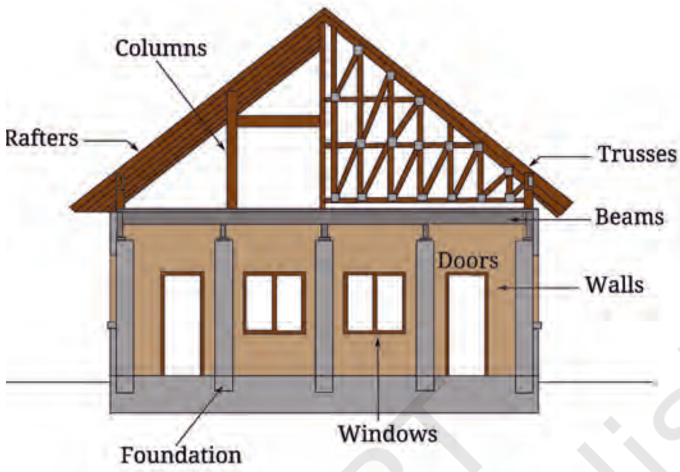

Figure 6.2: *Basic elements of buildings*

1. **Foundation:** A strong foundation is needed to increase the life of buildings, avoid cracks in the structure and safeguard against natural calamities like earthquakes. The depth of the foundation depends on the quantity of the load.
2. **Walls:** Different materials are used for making walls – bricks, stones, cement panels, wood, bamboo, etc.
3. **Beams:** Beams are the horizontal structural elements that carry the load of the slab (that is, upper floors) or roof. The size of the beam depends on the load of the slab.
4. **Columns:** The vertical structural elements supporting the beams are called 'columns'. They transfer the weight of the building to the foundation.
5. **Roofs:** The roof is the upper structural element of a building, designed and built using rafters (structural frameworks that connect different elements in a triangular fashion) or trusses (long sloping beams that make up the sloping roof) to support the load of the roof, and provide protection from weather, while completing the enclosure of the structure.
6. **Windows and doors:** These are structural openings in a building's walls to provide entry into the building, ventilation, natural light and aesthetic appeal.

6.2 Process Chart

6.2.1 Scoping work

Defining scope of work

Deciding the scope of the work means that decisions need to be taken regarding the following:

1. **What will be constructed:** This decision can be based on the need of the school or community. For example, a boundary wall, a shed, a ramp, a pavement or a wash basin. Figure 6.3 shows examples of some structures you could construct.

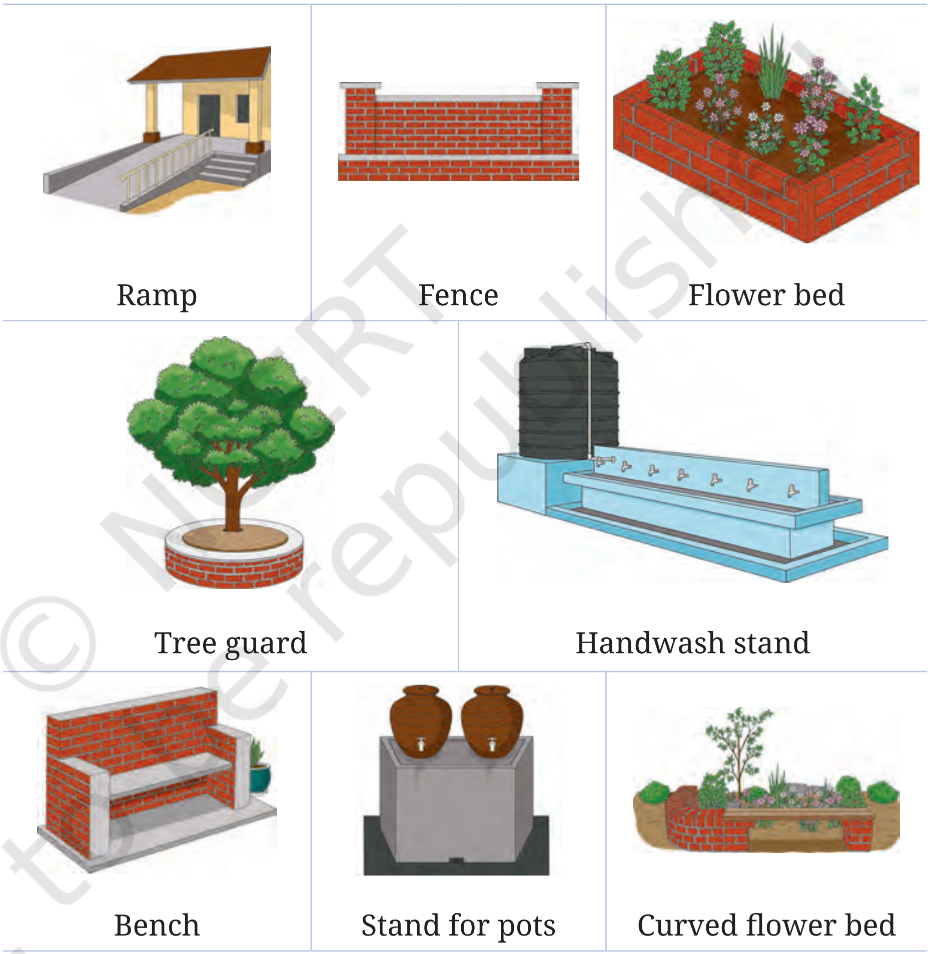

Figure 6.3: *Examples of structures you could construct*

2. **Material and resource availability:** Find out whether the materials and tools required for simple construction are available (for example, bricks, sand, cement, gravel, water or recycled materials). You should also consider the cost of these materials, and whether they can be safely managed and handled.

3. **What is useful:** Is the structure, which you have in mind, useful for the school or the community?
4. **Where will the construction take place:** You must decide the location where the structure will be built. It could be within or just outside the school. You must consider the space available, safety of the area, whether you can reach it easily, and also whether you can carry tools and materials to the site easily.

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1. What type of construction work will you take up in your school or community? Why?
2. Are there any specific considerations you will keep in mind?

6.2.2 Making a process chart

Making a process chart

Once you identify the structure and decide where to build it, you must list the key tasks, along with estimated dates by which they will be completed. Table 6.1 contains a sample.

Table 6.1: Sample process chart for construction

Tasks for construction	Dates	Responsibility
Preparing the construction drawing		
Marking the drawing on the field (line out)		
Preparation of foundation		
Brick work		
Curing (process to strengthen the construction)		
Plastering		
Finishing and painting		

6.3 Site visit

Before beginning work, you must visit a construction site in the surrounding area with a teacher and speak to practitioners. A practitioner can be a mason, civil contractor, civil engineer, architect or anyone else involved in actual design and construction. You can ask them about the processes, tools and materials to be used. You can also take their guidance for planning the work you will do.

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Use the pointers in Table 6.2 for discussion with practitioners and take notes.

Table 6.2: Pointers for observation during site visit

Points of observation/ discussions	Description
Tools and materials used	Materials used and their storage Tools used and their maintenance
Key processes	Key steps and their importance
Schedules	Frequency and timing of key tasks, if any
Safety protocols	Using appropriate tools, safety precautions, etc.
Quality criteria	Criteria for quality inputs, process and output
Use of technology	Digital tools/apps used

Think of any other points for observation while visiting the site. For example:

- What do the practitioners value the most about the work (for example, quality of construction, specific processes, pride in building something useful, etc.)?
- You can ask about possible challenges you may face during construction and how to overcome them.

Once you have returned from the site visit, work in groups to detail out the process chart further.

DID YOU KNOW?

Construction of houses depends on how the weight of the building is supported. On the basis of this, housing construction can be of two main types:

1. **Load bearing (wall-bearing) construction**

In this type of construction, the walls themselves carry the entire weight of the roof and upper parts of the building, and transfer it directly to the ground.

This method is commonly used for single-storey houses and areas where deep foundations are not needed. These houses are simple in design, cost less to build and are easy to repair (Figure 6.4).

Figure 6.4: *A single-storey house showing the load bearing construction*

2. **RCC (Reinforced Cement Concrete) construction**

In RCC construction, the building stands on a strong skeleton of columns and beams made of concrete and steel. Here, the load of the slab and floors is transferred to the ground through this frame, not through the walls.

Figure 6.5: *A multi-storey building showcasing RCC construction*

This type of construction is used for multi-storey buildings and structures that need to be very strong and earthquake-resistant (Figure 6.5).

6.4 Making a technical drawing of the selected structure

Technical drawing

It is possible that the person who is designing the construction (for example, an architect) is different from the person who is actually doing the work of construction (for example, civil engineer and mason). Technical drawings help communicate the exact details of work to be done.

Refer to Section 5.4.3 in Chapter 5 and draw the side view, plan and elevation of the structure you plan to construct.

CASELET

Students of Government High School decided to construct a ramp. They defined a scale and then made a technical drawing of the ramp (Figure 6.6).

Actual size	On paper dimension	Scale to be written as 1: XX (1 unit on paper: XX unit on the field)
10 m (1 m on field = 1 cm on paper)	10 cm	1: 100 (1 cm on paper = 100 cm on field)

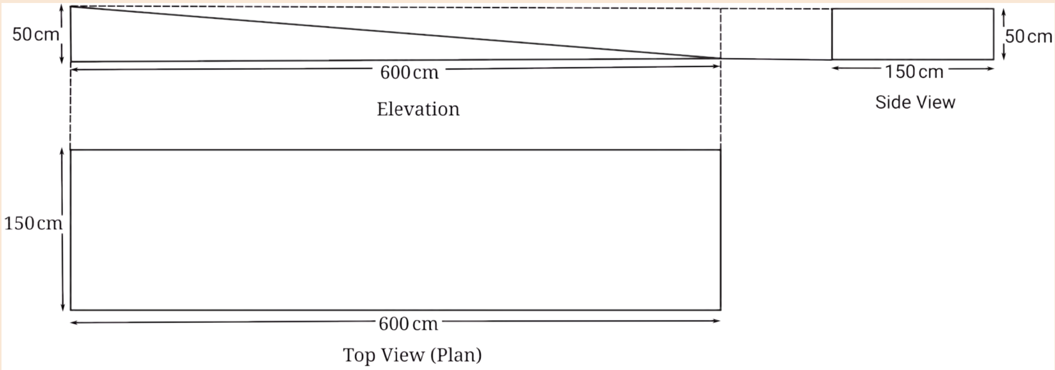

Figure 6.6: *Elevation, Side view and Plan of a ramp*

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Make a drawing to scale of the construction you want to take up. Show the side view, plan and elevation. Clearly mark the dimensions and the scale.

TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

In order to develop the technical drawing quickly and with accuracy, you can use computer-aided design (CAD) software. CAD also makes it easy to edit and rework the drawing.

Using CAD, you can:

1. Decide the exact measurements and scale of the structure;
2. Make 2D and 3D views of the structure;
3. Easily correct mistakes without erasing the entire drawing; and
4. Plan the use of materials and, therefore, reduce waste.

Further, CAD drawings can easily be shared with others as soft copies. You can search online to find suitable software as well as videos on making drawings using CAD.

Select tools and materials

6.5 Selecting materials

Table 6.3 lists the essential materials used for construction. You can select materials based on the type of structure you plan to construct in consultation with your teacher and/or the expert.

Table 6.3: Materials used for construction

Materials	Use	Safety note
Bricks	Used for building walls and partitions	Stack properly to avoid injury; handle gently – they can break or hurt someone if thrown around without care
Cement	Used as a binding material to hold bricks, sand and stones together	Avoid direct contact with skin and eyes; use gloves while handling; store material in a dry place
Sand	Mixed with cement to make mortar and concrete	Avoid inhaling dust; wear a mask while handling dry sand
Gravel	Mixed with cement and sand to make strong concrete	Lift sacks carefully to avoid back strain; handle with gloves
Water	Used for mixing cement and curing	Prevent water spills in areas used for standing and walking to avoid slipping
Paint	Used for finishing, and protecting walls and surfaces	Use in a well-ventilated area; wear mask to avoid inhaling fumes
Lime	Used in plaster and whitewashing	Avoid direct contact with eyes; wear gloves while handling

DID YOU KNOW?

Difference between mortar and concrete

Both mortar and concrete are important materials in construction. Mortar is used for binding other materials in a building (for example, 'gluing' bricks together, laying stone steps, installing bathroom tiles, etc.), while concrete is used for creating the actual structure and for supporting weight (for example, making a ramp, laying a foundation, making a fence, laying roads, etc.). Concrete 'reinforced' with steel bars (RCC) is used to support huge and heavy structures like skyscrapers (Figures 6.7 and 6.8).

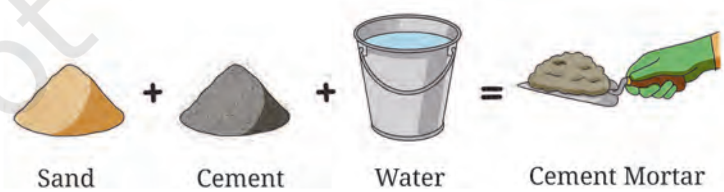


Figure 6.7: *Mortar is made up of sand, water, cement and lime – it is more flexible than concrete, as it allows for slight movement without cracking.*

Figure 6.8: *Concrete is made up of sand, water, cement, and gravel or crushed stones – it is a strong and durable building material that hardens over time, getting a stone-like quality.*

6.6 Selecting tools

Table 6.4 lists tools used for simple construction. Many technology and AI-based tools are also available to reduce manual efforts, for example, robots for tasks like brick-laying and demolition, light amplification by stimulated emission of radiation (laser) scanners to detect any faults in buildings.

Table 6.4: Tools used for construction

Tools	Use	Safety note
Trowel/Karni/ Thappa	Used for lifting, spreading, and smoothening cement or mortar	Hold firmly by the handle; avoid touching sharp edges and clean after use
Hoe	Used for loosening soil, mixing sand and cement on the ground, and levelling the surface	Use with proper grip; keep a safe distance from others while working
Wooden float	Used for smoothening and finishing the surface of plastered walls and floors	Use with dry hands for better grip; avoid slipping on wet surfaces
Metre tape	Used for measuring length, width, and height of walls and structures	If it is retractable, then retract slowly to avoid finger injury
Wire brush	Used for removing rust and dirt, or for smoothening rough surfaces	Use a face shield and gloves for protection against flying debris or wire filaments
Spirit level	Used to check whether a surface is perfectly horizontal (level) or vertical (straight)	Handle gently to avoid breaking the glass tube inside
Plumb bob	Used to check whether a wall or pillar is perfectly vertical	Do not swing near others; hold the string firmly

Water level tube	Long transparent rubber tube that is filled with water used to check horizontal level	Keep the ends secured to avoid spillage
Chisel	Used for cutting bricks, shaping stone, and removing extra cement or plaster	Hold firmly and keep fingers away from the cutting edge
Hammer	Used for striking the chisel, breaking bricks and fixing nails	Hold with a firm grip; do not strike on hard surfaces without control

6.7 Bill of Materials

Estimation

Bill of Materials (BoM) helps in estimating costs in advance and avoiding waste by ensuring only what is necessary is bought.

6.7.1 Estimating number of bricks required

Before making the bill of materials, you need to estimate the number of bricks required.

Bricks of different sizes and materials are available in the market, for example, clay bricks, fly ash bricks and concrete bricks. In India, the standard size of bricks is 190 × 90 × 90 mm (Figure 6.9). However, we need to consider 10 mm extra for the mortar we will apply to join bricks together. Hence, we should consider the brick size as 200 × 100 × 100 mm.

DID YOU KNOW?

Volume as basis for estimation of materials

When you build a structure out of bricks, imagine that you are filling a 3D space. To estimate how many bricks you need, you have to compare the total space of the structure with the space taken up by a single brick – this is the 'capacity' of the ramp to 'contain' bricks.

Every brick has a specific volume. By calculating the volume of the ramp, you can find its total 'capacity' to contain the required number of bricks. Some bricks may break or may need to be cut to the required size. To cater for this, engineers calculate the volume and add 10 per cent to the number of bricks.

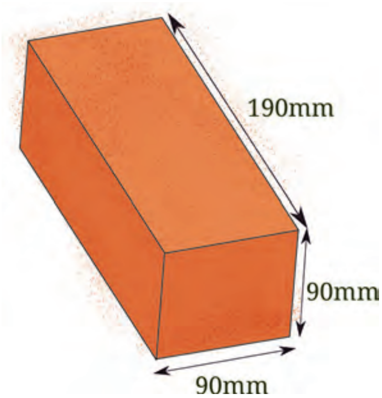


Figure 6.9: *Dimensions of a typical brick in India are 190 × 90 × 90 mm*

In India, bricks are typically made as per the specifications of the Bureau of Indian Standards (BIS), the body that maintains standards across products.

CASELET

Students of Government High School estimated the number of bricks required for constructing the ramp (Figure 6.10).

Volume of construction = Area of triangle (ramp) × width

Area of triangle (ramp) = ½ × 0.5 × 6 = 1.5 m^2

Width = 1.5 m

Volume = 1.5 × 1.5 = 2.25 m^3

By rule of thumb, 1 m^3 construction needs 500 bricks.

Therefore, 2.25 × 500 = 1125 bricks

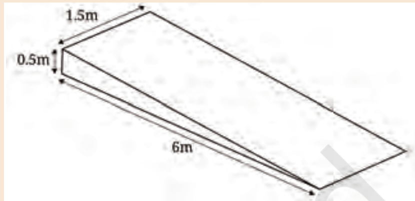


Figure 6.10: *Drawing of a ramp with dimensions*

To cater for any wastage, 10 per cent of the number of bricks is added to the total, so about 1200 bricks will need to be bought for making a structure of 1 m^3.

Please note that bricks are also available in non-standard sizes. Therefore, it is important to check the sizes of the bricks available, and then estimate the quantity.

Estimation

6.7.2 Estimate for mortar

A mixture of sand, cement and water is called mortar. Sand must be of good quality and it should be free of impurities like clay, dust, etc. Mortar is mainly used for plastering, brick work and similar tasks, as it creates a strong bond among the bricks. It is also used for the construction of walls, repair of construction and leakage, etc.

The recommended proportion for making mortar depends on the construction, as shown in Table 6.5.

Table 6.5: Ratio of cement and sand to make mortar for different kinds of construction work

Use of mortar	Ratio
Brick masonry	1:3
RCC	1:2
General repair and maintenance	1:3 to 1:6

By rule of thumb, typically to lay 100 standard bricks, approximately 5 kg to10 kg of cement is used for a ratio of 1:4

or 1:5 of cement and sand. Ask an expert about the ratio for the construction work you are doing.

Cost estimation and documentation

6.7.3 Preparing Bill of Materials

In addition to the actual cost of materials, the cost of labour must also be calculated. Cost of labour should also include the estimated time spent in doing the tasks.

CASELET

Students of Government High School prepared a Bill of Materials for construction of the ramp (Table 6.6).

Table 6.6: Bill of materials

Item	Quantity	Estimated cost (in ₹)	Remarks (if any)
Bricks	1200 × ₹ 10/brick	12000	
Cement	50 kg	400	
Sand	0.3 m^3 ~ 500 kg	500	
Water	100 L	-	

Cost of labour	Value (time spent in hours × hourly estimate × number of people)	Estimated cost (in ₹)	Remarks (if any)
Making foundation	1 hours × ₹ 50 × 2 people	100	
Marking layout	1 hours × ₹ 50 × 2 people	100	
Mixing cement and sand	2 hours × ₹ 50 × 2	200	
Brick laying	2 hours × ₹ 50 × 2 people	200	
Curing	10min/day × 28 days = 280 min ~ 4.6 hours × ₹ 50	230	
Total		**13,730**	

6.8 Minor repair and maintenance

Before you actually take up the selected construction work, you can practise making and using mortar for minor repair and maintenance work (Figure 6.11).

6.8.1 Carrying out minor repair

Identify the places in school/community/home, where minor maintenance of construction work is needed. For example, broken tiles, damaged walls, broken ramp, fencing, tree guard, small walls, etc.

TASK

Carrying out minor repairs

	**Step 1:** Mix one part cement and three parts sand.
	**Step 2:** Add water slowly to make a smooth, thick paste.
	**Step 3:** Clean the surface and lightly sprinkle water on it. Then apply the mortar on cracks, joints, tiles or uneven surfaces using a trowel.
	**Step 4:** Press, level and smoothen the applied area.

Figure 6.11: *Steps for carrying out minor repairs*

6.8.2 Curing and drying

Once you have applied mortar, it is necessary to keep it wet for 14–28 days. This process is called curing. It is necessary to complete the chemical reaction between water and mortar, which is important for 'bonding' materials together. This bonding process becomes stronger with curing. Generally, the process takes 28 days to complete. Hence, it is necessary to keep mortar wet after its application. If curing is stopped too early, then the structure becomes weak and cracks may appear, thereby reducing the strength of the structure. In case of minor repairs, you should leave the mortar to set for 24 hours. After 24 hours have passed, sprinkle water lightly for 14–28 days.

You can also perform curing by placing wet jute/gunny bags over the surface (especially if the water flows off sloped surfaces like a ramp) or sprinkling water through pipes, sprinklers or mugs to keep the surface continuously moist.

6.9 Making a structure

This section will discuss how to proceed with making the structure you have selected with the help of your teacher/expert.

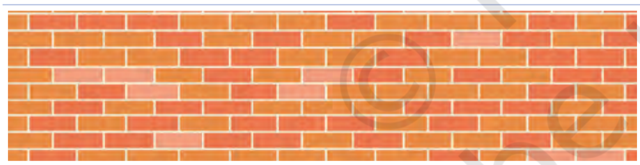
DID YOU KNOW?

Brick bonds

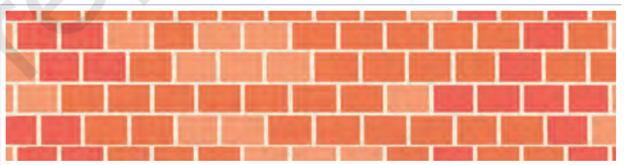
Different types of patterns are used for arranging and laying bricks using mortar. These patterns help make the structure strong and stable.

In Figure 6.12, you can see different methods of laying bricks for constructing a wall.

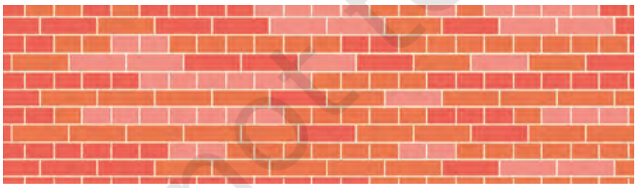
Use of tools and materials



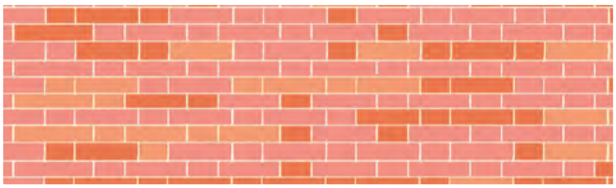
Stretcher bond: Length of brick facing front – used for boundary wall/partition with short span/wall of half brick thickness.



Header bond: Width of brick facing front – used for thick walls and also for curved walls.



English bond: Alternate rows of stretcher and header bonds – among strongest bonds mainly used for wall bearing structure.



Flemish bond: Alternate stretcher and header in each row.

Figure 6.12: *Different methods of laying bricks*

Constructing a structure

6.9.1 Preparation

The following steps are required to prepare for construction:

1. **Preparing ground:** As a first step, clean the construction site. Remove grass, loose soil, etc. Sprinkle water on the ground to settle any dust.
2. **Marking on ground:** Use a meter tape to mark the boundary of the structure; you can use a pointed stick to do so.
3. **Making the foundation:** Different types of construction need different kinds of foundations, as you read earlier in the chapter. In case of simple construction, a deep foundation is not necessary since the load is not high. However, it is necessary to ensure that the ground is flat and soil is firm for construction. Use a spirit level and water level tube to check if the ground is flat. If not, you will need to remove extra soil and flatten the surface. Spray water and let it dry to allow the soil to become firm.
4. **Putting a reference line for brick on ground:** Mark the layout of the structure on the levelled surface, using string or chalk lines to ensure the brickwork starts at the correct alignment, length and orientation before actual laying begins (Figure 6.13).

(a) (b)

Figure 6.13: *(a) Using reference lines for marking site of construction, and (b) marking lines for construction*

6.9.2 Preparing and laying bricks

Soak the bricks in water for a minimum of 6–12 hours. If you use dry bricks, then they will absorb water from the mortar.

Brick laying is done by placing bricks as required, and then applying mortar. Laying of the bricks must be done in the pattern of the selected bond, before applying mortar between them.

If you are making a ramp (or for that matter, any other structure that is not a perfect rectangle or square), you will need to cut bricks into the required shape and size so that they fit properly, ensuring structural strength and a smooth finish.

You should ensure that there are no gaps or edges protruding from the structure.

Figure 6.14: *Bricks are soaked in water before use*

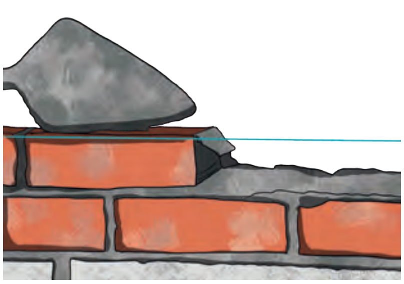


Figure 6.15: *Applying mortar while laying bricks*

Figure 6.16: *Cutting the brick using chisel and hammer*

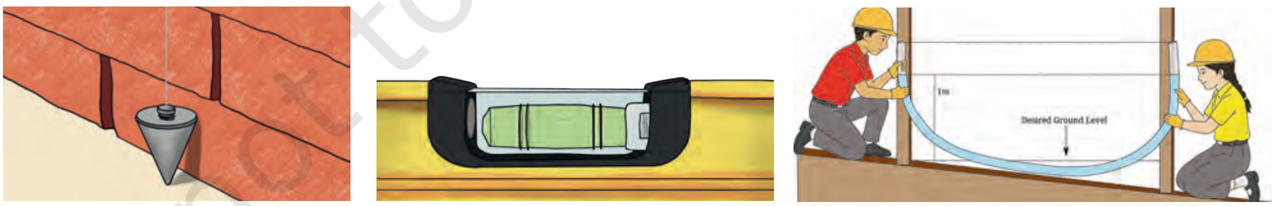
While there are advanced machines that cut the brick as per the desired shape, it needs precision and skill to handle the tool. However, you can cut the brick using a simple chisel and hammer under the supervision of an expert. You may refer Figures 6.14 to 6.16.

6.9.3 Ensuring alignment

Quality

Structures are designed to push the load straight down into the ground. If a structure is not perpendicular and parallel to the ground, or two points within it are not at the same height, gravity acts to weaken the structure.

Tools that are used to prevent this are the plumb bob (to ensure the structure is perpendicular to the ground), spirit level (to ensure the structure is parallel to the ground) and a water level tube (to ensure two points at a distance are at the same height) (Figure 6.17).



(a) (b) (c)

Figure 6.17: *(a)Plumb bob must hang parallel to the ground; (b) bubble in spirit level tube must be at the centre; and (c) level in water level tube must be the same at all points of the structure*

Finishing

6.9.4 Curing and plastering

After completing the brickwork, the next step is plastering.

Plastering is the process of applying a smooth or textured coating of plaster to a wall. This is done to increase the durability of the construction, provide protection from moisture and heat, and to make the structure look neat and attractive. Specific tools are used for plastering (Figure 6.18).

The following steps need to be followed for plastering the wall:

1. **Preparation of surface**: Use a wire brush to remove any loose debris on the wall. Also lightly dampen the wall with water.
2. **Making cement plaster for plastering:** Do not make the paste for plastering in large quantities. Mix the material and add water as required.

Generally, the proportion of cement to fine sand is taken as 1:6 for external plastering and 1:4 for interior plastering. You can add water slowly to the mixture to form the paste with the required consistency. The mix should be thick enough to not slide off the wall.

Trowel/Thappa/Karni: It is used to apply and flatten the plaster.

Float: It is used to flatten the plaster.

Wire Brush: For cleaning

Figure 6.18: *Tools used for plastering*

3. **Apply plaster:** First apply a base coat of plaster. Allow it to set, that is, become dry and hardened. Generally, the thickness of plaster is 12–15 mm. Apply a second coat of plaster to complete the plastering (Figure 6.19).

Figure 6.19: *Applying plaster on the wall*

Figure 6.20: *Plastering of a wall*

4. **Finishing:** Use a finishing tool and trowel to get a smooth texture (Figure 6.20).
5. **Drying and curing:** Carry out curing for several days. Keep the surface wet by spraying water regularly.

The construction of the basic structure is now complete except for a few finishing touches.

CHECK YOUR UNDERSTANDING

Think about the work you did. What did you enjoy? Did you face any challenges? If yes, describe them and what you did to overcome them.

6.10 Finishing the product

Finishing

Once plastering and curing have been completed, the construction work is almost complete. However, some finishing touches will make your work look professional.

Painting helps in making walls look neat and attractive. Besides aesthetics, painting helps in increasing the life of the construction by preventing leakages. It also helps in pest control.

Different types of paints are available in the market. Visit a hardware shop and select the most suitable paint based on your budget and choice.

Table 6.7 contains a summary of different types of paints commonly used in India, along with their applications and advantages.

Table 6.7: Different types of paints used in India

Type of Paint	Applications	Advantages
Whitewash (Lime wash)	Houses, temporary structures, ceilings, low-cost housing	Very cheap, antibacterial properties, reflects heat, eco-friendly
Distemper	Interior walls and ceilings in low-cost buildings	Economical, easy to apply, available in many colours, breathable surface
Cement Paint	Exterior walls, boundary walls, basements, damp areas	Water-resistant, prevents fungus/ mildew, economical for large surfaces, durable in outdoor use

CASELET

Students of Government High School decided that the most suitable finishing paint for the ramp is cement paint, since it is water-resistant and prevents slipperiness. They followed the process below to apply cement paint on the ramp (Figure 6.21):

1. **Surface cleaning:** First, clean the ramp surface properly. Remove dust, loose cement, sand, grease and any old paint using a brush, broom or water wash. A clean surface helps the paint stick better.
2. **Wetting the surface:** Before applying cement paint, sprinkle clean water on the ramp. Cement paint must be applied on a slightly wet surface for proper bonding.

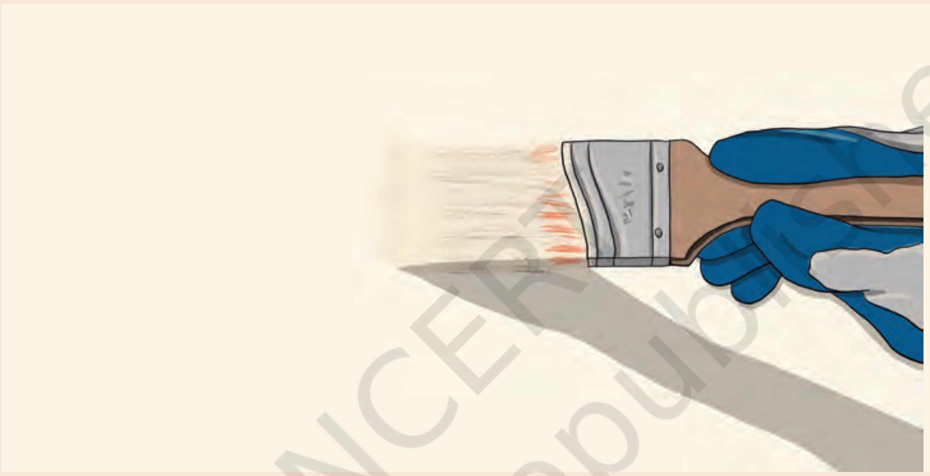


Figure 6.21: *Applying first coat of paint on side of ramp*

3. **Preparing the paint:** Mix the cement paint with water as per the instructions given on the packet. Stir well to get a smooth mixture without lumps.
4. **Applying the first coat:** Apply the first coat using a brush or roller, starting from one end of the ramp and moving evenly across the surface. Apply in one direction for a smooth finish.
5. **Curing the first coat:** Allow the first coat to dry for at least 24 hours. Lightly sprinkle water on the painted surface during this time for proper setting.
6. **Applying the second coat:** Once the first coat is dried for at least 24 hours, apply the second coat in the same manner. This improves colour, durability and water resistance.
7. **Final drying and safety check:** Let the ramp dry for 48–72 hours before use. Put up a 'Wet Paint – Do Not Use' sign to avoid accidents.
8. **Post-construction cleaning:** Cleaning the place around a construction site is important for safety, and a neat and attractive appearance. Remove trash from the site and store unused material in the assigned place.

PORTFOLIO

Maintain a record of the work done by you. Take photographs, if possible, at each step of construction. Some sample images of work done by students in other schools are given in Figure 6.22.

Figure 6.22: *Structures built by students*

6.11 While constructing a structure

1. Before beginning any work, it is necessary to learn through observation. This can be done by visiting a construction site, online search or interviewing experts in the area.
2. Preparation of ground of construction is an important step. Take guidance of experts to decide the depth of the foundation, depending on the weight of the structure and soil type.
3. Use a water level tube to ensure that two distant points of the construction are at the same level and a spirit level to check that all surfaces are level with the ground. Also, use a plumb bob to check that the construction is perpendicular to the ground.
4. Always soak the bricks in water for 6–12 hours before using them. Use correct proportion of sand, cement and water for making mortar and sand, cement, gravel, and water for preparing concrete.
5. Spray water on the construction work after applying concrete or mortar for a minimum of 14–28 days.
6. Do not allow the construction to dry out. This process of applying water to new construction is called curing. After construction, clean all the surroundings for any debris, unused sand and cement.
7. Painting the constructed structure will not only make it attractive but also increase its durability.

6.12 Assess your learning

1. Look around your school or neighbourhood, and identify any one structure. Identify the basic structural elements (foundation, walls, roof, beams or columns) that you can observe and describe their functions.
2. What kind of construction has been used for your house or school (Load-bearing construction or RCC construction)? Write the reasons for selecting particular types of construction.
3. Suppose you are asked to construct a small boundary wall in your school. List the major steps you would include in the process chart and explain why sequencing of steps is important.
4. Identify three safety rules that must be followed at a construction site. Explain how ignoring any one of them could lead to an accident.
5. Cement was left exposed in the room during the rainy season for many months. What might happen to cement and why ?
6. Often, during summer season, especially during a drought year, the Government bans construction activities. Why do you think this is done?
7. A student is laying bricks to make a boundary wall and decides to skip using the plumb bob to save time, claiming they can 'see' if the wall is straight. What is the specific technical risk of relying on visual estimation instead of a plumb bob? If the wall is even slightly 'out of plumb' (not perfectly vertical), how might this affect the building's safety over the years?
8. Of the tasks that you did, which did you enjoy the most? Which did you enjoy the least? Give examples of what went well and what did not go well. What would you do differently next time?
9. Give examples of how you can apply your learnings in a real-life situation.